

- (a) $\frac{1}{6}$ (b) $2\frac{7}{12}$
 (c) $15\frac{1}{2}$ (d) $21\frac{1}{2}$
80. $\frac{7\frac{1}{2} - 5\frac{3}{4}}{3\frac{1}{2} + ?} \div \frac{\frac{1}{2} + 1\frac{1}{4}}{1\frac{1}{5} + 3\frac{1}{2}} = 0.6$
 (a) $4\frac{1}{3}$ (b) $4\frac{1}{2}$
 (c) $4\frac{2}{3}$ (d) None of these
81. Which of the following pairs of fractions adds up to a number greater than 5?
 (a) $\frac{5}{3}, \frac{3}{4}$ (b) $\frac{7}{3}, \frac{11}{5}$
 (c) $\frac{11}{4}, \frac{8}{3}$ (d) $\frac{13}{5}, \frac{11}{6}$
82. $5 - \left[\frac{3}{4} + \left\{ 2\frac{1}{2} - \left(0.5 + \frac{1}{6} - \frac{1}{7} \right) \right\} \right]$ is equal to
 (a) $1\frac{19}{84}$ (b) $2\frac{61}{84}$
 (c) $2\frac{23}{84}$ (d) $2\frac{47}{84}$
83. If $[p]$ means the greatest integer less than or equal to p , then $\left[-\frac{1}{4} \right] + \left[4\frac{1}{4} \right] + [3]$ is equal to:
 (a) 4 (b) 5
 (c) 6 (d) 7
84. If $x = -3$, then $x^3 - x^2 - x$ will be equal to (U.P.G.I.C., 2009)
 (a) -33 (b) -27
 (c) 15 (d) 54
85. Let $R = gS - 4$, when $S = 8$ and $R = 16$. If $S = 10$, then R is equal to
 (a) 11 (b) 14
 (c) 20 (d) 21
86. If $x = 5$, $y = 3$ and $z = 2$, then $\frac{x(y-z)}{y(x+y+z)} = ?$
 (a) $\frac{1}{6}$ (b) $\frac{1}{30}$
 (c) 1 (d) 5
87. If $a = 7$, $b = 5$, then the value of $a^3 - b^3 + 3a^2b$ is
 (a) 218 (b) 307
 (c) 735 (d) 953
88. $-7m - [3n - \{8m - (4n - 10m)\}]$ simplifies to (R.R.B., 2006)
 (a) $11m - 5n$ (b) $11m - 7n$
 (c) $11n - 7m$ (d) $13n - 11m$

89. If $a + 2b = 6$ and $ab = 4$, then what is $\frac{2}{a} + \frac{1}{b}$? (M.B.A., 2006)
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{3}{2}$ (d) 2
 (e) $\frac{5}{2}$
90. If $\frac{a}{3} = \frac{b}{4} = \frac{c}{7}$, the value of $(a + b + c) \div c$ is
 (a) $\frac{1}{7}$ (b) $\frac{1}{2}$
 (c) 2 (d) 7
91. The value of $2a^3 - [3a^3 + 4b^3 - \{2a^3 + (-7a^3)\} + 5a^3 - 7b^3]$ is (Teacher's Exam, 2007)
 (a) $-11a^3 + 3b^3$ (b) $7b^3 + 3a^3$
 (c) $11a^3 - 3b^3$ (d) $-(11a^3 + 3b^3)$
92. If $\frac{x}{y+z} = a$; $\frac{y}{z+x} = b$ and $\frac{z}{x+y} = c$, then $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c}$ is equal to
 (a) 2 (b) 3
 (c) 6 (d) $x + y + z$
93. If $y = (x + 3)^2$, then $(-2x - 6)^2$ is equal to
 (a) $-4y^2$ (b) $-2y^2$
 (c) $-4y$ (d) $2y$
 (e) $4y$
94. Which among the following is the correct value of $(x + a)(x + b)$? (R.R.B., 2007)
 (a) $x^2 + abx + ab$ (b) $x^2 - (a + b)x + ab$
 (c) $x^2 + (a + b)x + ab$ (d) $x^2 + (a - b)x + ab$
95. The value of $\left[1 + \frac{1}{x+1} \right] \left[1 + \frac{1}{x+2} \right] \left[1 + \frac{1}{x+3} \right] \left[1 + \frac{1}{x+4} \right]$ is
 (a) $\frac{x+5}{x+1}$ (b) $\frac{x+1}{x+5}$
 (c) $1 + \frac{1}{x+5}$ (d) $\frac{1}{x+5}$
96. If $(a - b)$ is 6 more than $(c + d)$ and $(a + b)$ is 3 less than $(c - d)$, then $(a - c)$ is
 (a) 0.5 (b) 1
 (c) 1.5 (d) None of these
97. If $\frac{a}{b} = \frac{4}{3}$, then $\frac{3a+2b}{3a-2b} = ?$ (M.B.A., 2007)
 (a) -1 (b) 3
 (c) 5 (d) 6